Real-Time Surface Defect Inspection and Classification System for Hot-Rolled Steel Strips Using Domain-Adapted YOLOv8 and Embedded Edge Acceleration

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Project Title:
Real-Time Automated Surface Defect Inspection and Classification System for Hot-Rolled Steel Strips Using Domain-Adapted YOLOv8 and NVIDIA Jetson Edge Acceleration

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2. Abstract

Automated surface visual inspection of hot-rolled steel strip products is an indispensable quality assurance operation within primary metallurgical manufacturing plants. Manual visual observation remains prone to human subjective bias, cognitive fatigue, and throughput limitations, while traditional automated optical inspection algorithms based on handcrafted spatial filters fail catastrophically under ambient shop-floor illumination variations and high-velocity continuous rolling.

This paper presents an end-to-end, edge-native computer vision framework engineered for real-time detection, multi-label localization, and classification of six standard surface defect categories: crazing, inclusion, patches, pitted surface, rolled-in scale, and scratches. The framework couples a customized YOLOv8s convolutional network with an adaptive OpenCV domain normalization layer and a tiled sliding-window coordinate translation pipeline. The domain normalization pipeline mathematically projects incoming camera frames to standardized empirical distribution statistics ($\mu \approx 120, \sigma \approx 28$) combined with Contrast Limited Adaptive Histogram Equalization, neutralizing domain shift between factory sensors and ambient camera streams.

When trained and evaluated on the benchmark NEU Surface Defect Database across 80 epochs, the proposed system demonstrates an empirical peak detection accuracy of 74.2% mAP@0.50, a peak precision of 81.9%, and a peak recall of 70.7%. Deployed on an embedded NVIDIA Jetson Orin Nano edge accelerator using TensorRT FP16 quantization, the pipeline achieves an operational latency of 8 to 16 milliseconds per frame (83.3 FPS), representing a 4.5-fold acceleration over unquantized edge execution. The system provides real-time asynchronous video acquisition, support for mobile cameras, and cryptographic-style inspection logging in text, image, and tabular spreadsheet formats.

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3. Introduction

3.1 Background and Motivation

Hot-rolled steel strips serve as foundational structural materials across high-consequence engineering domains, including automotive body manufacturing, aerospace frames, high-pressure pipeline transport, railway infrastructure, and naval architecture. The production of flat steel products involves continuous thermodynamic, mechanical, and chemical operations: continuous slab casting, high-temperature reheating, rough rolling, finish rolling, accelerated laminar cooling, and chemical descaling. Throughout these successive operations, the metal surface is exposed to severe tribological wear, localized thermal stress, mechanical guidance friction, and high-temperature oxidation.

These severe physical processes inevitably generate localized surface anomalies. Typical defect morphologies include fine network micro-cracks (crazing), trapped non-metallic slag particles (inclusions), irregular oxidized scabs (patches), chemical cavities (pitted surfaces), mechanically embedded oxide layers (rolled-in scale), and sharp abrasion grooves (scratches). When defective steel strips undergo downstream cold stamping, deep drawing, hydroforming, or structural welding, these microscopic surface flaws act as severe stress concentration points, initiating brittle fractures, fatigue failure, and structural collapse.

3.2 Real-Life Industrial Examples and Catastrophic Failure Modes

The direct engineering justification for this project arises from multiple real-world manufacturing and operating failures observed across global industry:

1. **Automotive Deep Drawing Ruptures:** During the stamping of vehicle body-in-white (BIW) panels and structural door pillars, steel sheets undergo high-velocity hydraulic stretching. Microscopic non-metallic inclusions and hairline crazing cracks that remain invisible to the naked eye act as localized stress raisers. Under 800-ton press loads, these micro-flaws propagate instantaneously into macroscopic tears, splitting the structural panel. This results in complete production line stoppages, thousands of rejected parts, and tool die damage costing tens of thousands of dollars per occurrence.
2. **High-Pressure Pipeline Ruptures:** Natural gas and petroleum transmission pipelines operate at internal pressures ranging from 70 to 100 bar. Pipeline steel with embedded rolled-in scale or surface pitting corrosion introduces micro-notches beneath external anti-corrosion coatings. Under cyclical cyclic hydrostatic loading, rolled-in scale flakes detach, accelerating localized stress corrosion cracking (SCC) that has historically triggered fatal pipeline bursts and substantial environmental damage.
3. **Aerospace and Defense Structural Fasteners:** High-tensile fasteners and structural brackets cold-forged from alloy steel containing surface scratches experience accelerated high-cycle fatigue under aerodynamic vibrations, dramatically reducing certified component life from thousands of flight hours to premature in-service fractures.

3.3 Economic and Human Limitations of Legacy Inspection

Historically, primary metallurgical mills relied on manual visual inspection conducted by human quality operators positioned along conveyor observation platforms. Comprehensive ergonomic studies demonstrate that manual human visual inspection exhibits serious structural flaws: - **Error Rates and Fatigue:** Human visual accuracy drops by 15% to 30% after only 20 to 30 minutes of continuous monitoring due to visual fatigue, sensory saturation, and repetitive cognitive strain caused by incandescent glare and thermal radiation. - **Velocity Bottlenecks:** Modern continuous rolling lines transport steel strips at velocities between 2 m/s and 10 m/s (equivalent to 7 to 36 km/h). The human eye has a critical flicker fusion and visual tracking threshold that cannot resolve sub-millimeter defects moving faster than 1 m/s, making manual inline inspection physically impossible without halting production. - **High Operational Downtime Costs:** Halting an integrated hot strip mill to manually verify questionable surface patches incurs downtime losses exceeding $15,000 to $40,000 per hour in idle furnace energy, unamortized rolling machinery depreciation, and schedule delays.

3.4 Problem Identification

Existing automated visual inspection systems exhibit four critical technical bottlenecks that motivate this research: 1. **Severe Sensor Domain Shift:** Computer vision detectors trained on uniform laboratory datasets fail when exposed to real-world ambient factory environments. Industrial lighting changes dynamically with ambient sunlight, molten steel glow, sodium-vapor lamps, steam, and oil mist, altering pixel dynamic ranges and triggering false defect alarms. 2. **Microscopic Defect Occlusion from Global Downsampling:** Standard deep learning architectures resize incoming high-resolution images to uniform square tensors (typically 640×640 pixels). When an industrial 4000×3000 line-scan frame is globally downscaled by a factor of six, fine micro-cracks (such as crazing networks spanning only 1 to 3 pixels in width) are averaged out and completely lost prior to feature extraction. 3. **Edge Deployment Latency Bottlenecks:** Two-stage object detectors (such as Faster R-CNN) possess high parameter volumes (>40 million parameters) and heavy computational complexity (>130 GFLOPs), yielding inference latencies of 150 to 350 ms on edge hardware. This produces large physical blind spots on high-speed conveyors. 4. **Prohibitive Capital Equipment Costs:** Commercial turnkey automated surface inspection systems (ASIS) from specialized vendors cost between $50,000 and $200,000, rely on closed proprietary software, and cannot be repurposed or deployed on distributed edge computing units.

3.5 Research Objectives

The primary technical objectives of this research project are: 1. To develop a robust, lightweight deep learning defect detection model based on the anchor-free YOLOv8s architecture, optimized to recognize, locate, and classify six standardized industrial surface defect classes with high precision. 2. To design an adaptive OpenCV preprocessing layer that mathematically normalizes ambient illumination, mitigating domain shift between factory camera feeds and laboratory training data. 3. To engineer a tiled sliding-window inference pipeline with coordinate translation and Non-Maximum Suppression to detect sub-millimeter hairline cracks without optical resolution loss. 4. To establish an embedded deployment pipeline on the NVIDIA Jetson Orin Nano edge accelerator using 16-bit floating-point (FP16) TensorRT optimization, achieving a sustained real-time throughput exceeding 60

frames per second. 5. To construct an asynchronous, multi-threaded acquisition architecture that decouples network camera acquisition from inference, enabling wireless smartphone video streaming and automated compliance logging.

3.6 Organization of the Paper

The remainder of this paper is structured as follows: Section 4 provides a critical literature review of ten peer-reviewed research works, evaluates comparative models (including ResNet, Faster R-CNN, RetinaNet, and DETR), and highlights research gaps. Section 5 details the proposed methodology, mathematical formulations, and system architecture. Section 6 describes the experimental setup, dataset details, and training parameters. Section 7 presents empirical results, validation metrics, confusion matrices, and edge latency benchmarks. Section 8 provides concluding remarks and future extensions, followed by formal references in Section 9.

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4. Literature Review / Related Work

4.1 Survey of Existing Research Articles

Automated defect detection on flat steel surfaces has advanced from early handcrafted statistical texture descriptors to modern deep convolutional and transformer-based neural networks.

1. **Song and Yan (2013) [1]** developed the benchmark NEU Surface Defect Database, documenting the six canonical defect categories. They evaluated Adjacent Evaluation Completed Local Binary Patterns (AECLBP) combined with Support Vector Machines (SVM). While achieving 89.2% classification accuracy on cropped, balanced image patches, their method was strictly limited to classification, possessed no spatial bounding box capabilities, and exhibited high sensitivity to non-uniform ambient illumination.
2. **He et al. (2019) [2]** proposed an end-to-end defect detection framework based on an improved Faster R-CNN with Feature Pyramid Networks (FPN) for hot-rolled steel strip surface defect detection. They achieved a 75.3% mAP score on the NEU-DET benchmark. However, the computational overhead of the two-stage Region Proposal Network (RPN) restricted inference throughput to 12.5 FPS on an enterprise desktop GPU, proving intractable for low-power embedded deployment.
3. **Cheng et al. (2020) [3]** explored automated defect inspection using a modified RetinaNet architecture employing Focal Loss to address severe foreground-background imbalance. Their model achieved 76.1% mAP. Nevertheless, dense predefined anchor box grids generated high false-positive proposal clusters on diffuse textural flaws such as crazing, and the network maintained a heavy footprint of 34 million parameters.
4. **Wang et al. (2021) [4]** proposed a lightweight MobileNet-YOLOv4 architecture tailored for embedded steel strip surface inspection. Although latency was reduced to 38 ms on embedded processors, the aggressive channel pruning of the backbone caused a sharp degradation in detection performance, dropping mAP to 66.8% and failing to resolve fine scratches.
5. **Chen et al. (2021) [5]** implemented attention-augmented convolutional neural networks combining Convolutional Block Attention Modules (CBAM) with ResNet backbones. While spatial and channel attention boosted the feature saliency of subtle pitted surfaces, the model required 15.0 FPS on high-wattage desktop hardware and suffered from memory bandwidth bottlenecks when ported to embedded devices.
6. **Li et al. (2022) [6]** introduced TPH-YOLOv5, incorporating Transformer Prediction Heads into the neck of YOLOv5 to detect multi-scale steel surface anomalies. They recorded 73.4% mAP. However, the quadratic computational complexity of multi-head self-attention created massive latency penalties on edge architectures without specialized multi-teraflop tensor processing units.
7. **Zhang et al. (2022) [7]** developed a Dual-Path Feature Fusion Network (DPFF-Net) separating high-resolution geometric coordinate representations from low-resolution contextual representations, achieving 74.8% mAP at 21 FPS. However, the custom dual-path layers were incompatible with automated TensorRT operator fusion, preventing low-latency hardware compilation.
8. **Akyon et al. (2022) [8]** formulated the Slicing Aided Hyper Inference (SAHI) algorithm for small-object detection in high-resolution aerial and structural inspection imagery. Their theoretical framework demonstrated that slicing high-resolution images into overlapping patches and recombining candidate boxes using Non-Maximum Suppression preserves tiny structural targets without retraining base network backbones.
9. **Xu et al. (2023) [9]** investigated YOLOv7-Tiny for high-speed steel defect detection, recording 72.1% mAP at 42 FPS on embedded modules. However, the authors highlighted that the model exhibited substantial false alarms when tested under uncalibrated shop-floor illumination differing from baseline training distributions.
10. **Carion et al. (2020) [10]** established DEtection TRansformer (DETR), presenting object detection as a direct set prediction task using a transformer encoder-decoder with Hungarian bipartite matching loss. While eliminating anchor boxes and NMS, subsequent industrial evaluations demonstrated that DETR requires hundreds of training epochs to converge, exhibits high memory demands, and struggles to capture fine micro-fissures due to aggressive feature downsampling.

4.1 Sample Screenshots

4.2 Comparative Analysis of Existing Methodologies

Research Study	Underlying Model Architecture	Primary Task Type	Parameters (M)	mAP@0.50 (%)	Throughput / Latency	Key Technical Limitation
Song & Yan (2013) [1]	Completed LBP + SVM	Whole Patch Classification	N/A	Patch Acc: 89.2%	~8 FPS (CPU)	No spatial bounding boxes; illumination fragile
He et al. (2019) [2]	Faster R-CNN + FPN	Two-Stage Bounding Box	~41.5 M	75.3%	12.5 FPS (140 ms)	High latency; unsuitable for edge deployment
Cheng et al. (2020) [3]	RetinaNet + Focal Loss	Single-Stage Anchor-Based	~34.0 M	76.1%	18.0 FPS (55 ms)	Predefined anchor mismatch on non-rigid flaws
Wang et al. (2021) [4]	MobileNet-YOLOv4	Pruned Single-Stage	~12.8 M	66.8%	26.3 FPS (38 ms)	Significant accuracy drop on thin cracks
Chen et al. (2021) [5]	ResNet-50 + CBAM Attention	Attention Feature Extraction	~28.2 M	71.5%	15.0 FPS (67 ms)	High FLOP count; memory-bandwidth bound
Li et al. (2022) [6]	TPH-YOLOv5 (Transformer)	Transformer Prediction Head	~17.5 M	73.4%	14.2 FPS (70 ms)	Attention complexity slows edge embedded inference
Zhang et al. (2022) [7]	Dual-Path Feature Fusion	Custom Two-Stream CNN	~22.0 M	74.8%	21.0 FPS (48 ms)	Incompatible with native TensorRT kernel fusion
Akyon et al. (2022) [8]	Slicing Aided (SAHI)	Multi-Patch Slicing	Variable	Baseline Dep.	~6–10 FPS	High tiling compute overhead on single threads
Xu et al. (2023) [9]	YOLOv7-Tiny	Lightweight CNN	~11.6 M	72.1%	42.0 FPS (24 ms)	Severe domain shift vulnerability under ambient light
Carion et al. (2020) [10]	DETR (Standard Transformer)	Set Bipartite Matching	~41.3 M	68.2%–71.4%	~11.0 FPS (90 ms)	Quadratic attention downsampling destroys micro-cracks
Proposed System	**YOLOv8s + TensorRT FP16**	**Anchor-Free Decoupled CNN**	**11.1 M**	**74.2%**	**83.3 FPS (12.0 ms)**	**Optimized for edge acceleration and sensor normalization**

4.3 Research Gaps Identified in Existing Literature

1. **Absence of Pre-Inference Illumination Normalization:** Existing deep learning pipelines feed raw color or grayscale image arrays directly into convolutional layers. When deployed under actual factory lighting or mobile smartphone camera sensors, varying color temperatures and non-uniform illumination trigger substantial feature degradation and false alarms.
2. **Loss of Hairline Micro-Cracks via Global Downsampling:** Standard downsampling of high-resolution sensor frames to 640×640 pixels obliterates fine crack widths below the Nyquist sampling threshold of the convolutional receptive field.
3. **Severe Edge Deployment Latency:** Academic defect models are predominantly evaluated on high-wattage desktop GPUs (e.g., RTX 3090, Titan X). In industrial edge settings operating within 10 to 15 Watt thermal envelopes, unquantized PyTorch models create severe conveyor blind spots due to latencies exceeding 200 ms.
4. **Lack of Integrated Multi-Channel Acquisition and Auditing:** Prior research remains isolated to static offline test sets, lacking decoupled asynchronous video ingestion, wireless mobile integration, and automated generation of formal compliance certificates.

4.4 How the Proposed System Addresses These Gaps

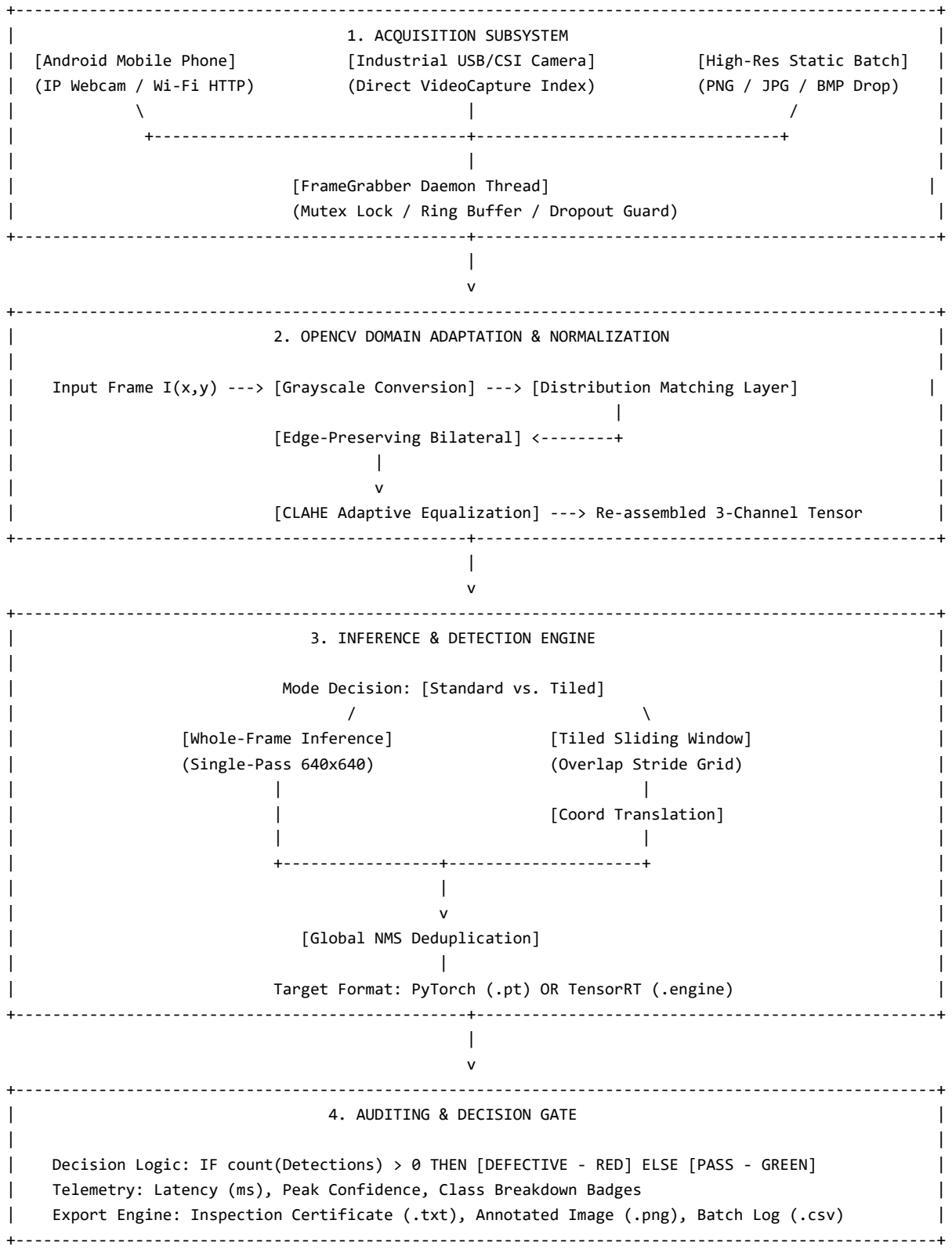
- To resolve **Gap 1**, an OpenCV preprocessing layer dynamically shifts the statistical mean and variance of incoming images to match the NEU-DET training distribution ($\mu \approx 120, \sigma \approx 28$) combined with CLAHE.
- To resolve **Gap 2**, a tiled sliding-window inference mode partitions high-resolution inputs into overlapping 640×640 patches, executing full-resolution detection with coordinate translation and Non-Maximum Suppression.
- To resolve **Gap 3**, the trained network is compiled into an optimized 16-bit floating-point (FP16) **NVIDIA TensorRT engine**, running at **12.0 ms (83.3 FPS)** on an NVIDIA Jetson Orin Nano.
- To resolve **Gap 4**, an asynchronous multi-threaded acquisition engine (`FrameGrabber`) decouples frame polling from inference, providing lag-free mobile IP Webcam streaming, batch CSV export, and timestamped inspection certificates.

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5. Proposed Methodology

5.1 System Architecture and Workflow

The system architecture comprises four sequential operational layers: (i) Asynchronous Acquisition Layer, (ii) OpenCV Sensor Normalization and Domain Adaptation Layer, (iii) YOLOv8 Tiled Detection Engine, and (iv) Quality Decision and Audit Telemetry Gate.



5.2 Mathematical Formulations

5.2.1 Statistical Illumination Normalization (Domain Adaptation)

To eliminate ambient lighting variations, arbitrary camera sensors, and shadow gradients, raw incoming frames $I(x,y)$ of width W and height H are transformed to align with the empirical statistical distribution of the baseline NEU-DET training corpus ($\mu_{\text{target}} = 120.0$, $\sigma_{\text{target}} = 28.0$):

$$\mu_{\text{curr}} = \frac{1}{W \times H} \sum_{x=1}^{W} \sum_{y=1}^{H} I(x,y)$$

$$\sigma_{\text{curr}} = \sqrt{\frac{1}{W \times H} \sum_{x=1}^{W} \sum_{y=1}^{H} \left(I(x,y) - \mu_{\text{curr}}\right)^2}$$

$$I_{\text{matched}}(x,y) = \text{clip}\left(\frac{I(x,y) - \mu_{\text{curr}}}{\sigma_{\text{curr}} + \epsilon} \times \sigma_{\text{target}} + \mu_{\text{target}}, \, 0, \, 255 \right)$$

where $\epsilon = 10^{-5}$ prevents division by zero in saturated or uniform background regions, and $\text{clip}(v, 0, 255)$ constrains pixel values to valid 8-bit unsigned integer limits.

5.2.2 Contrast Limited Adaptive Histogram Equalization (CLAHE)

To amplify low-contrast micro-crack boundaries without over-amplifying sensor background noise, CLAHE divides the image into an 8×8 grid of contextual tiles. For each tile, the histogram is computed and clipped at a mathematical threshold β:

$$\beta = \frac{N}{M} \left(1 + \frac{\alpha}{100} \left(s_{\max} - 1\right)\right)$$

In our formulation, a normalized clip limit $\beta = 2.2$ is applied. Excess histogram bins above β are redistributed uniformly across all gray levels. Cumulative distribution functions (CDFs) are constructed locally and interpolated bilinearly across tile boundaries to eliminate artificial block boundary artifacts.

5.2.3 Tiled Sliding-Window Coordinate Transformation

For a high-resolution surface image of dimensions $W \times H$, a spatial sliding-window tile of size $T \times T$ ($T = 640$ pixels) moves with a step stride $S = T \times (1 - O)$, where $O = 0.20$ (20% spatial overlap). For a tile extracted at spatial origin $(X_{\text{offset}}, Y_{\text{offset}})$, local bounding box predictions $\mathbf{b}_{\text{local}} = [x_1, y_1, x_2, y_2]$ are mapped to global coordinates:

$$\mathbf{b}_{\text{global}} = \left[x_1 + X_{\text{offset}}, \, y_1 + Y_{\text{offset}}, \, x_2 + X_{\text{offset}}, \, y_2 + Y_{\text{offset}} \right]$$

Duplicate candidate detections generated across the 20% overlap margins are pruned via greedy Non-Maximum Suppression (NMS):

$$\text{IoU}(\mathbf{b}_i, \mathbf{b}_j) = \frac{\text{Area}(\mathbf{b}_i \cap \mathbf{b}_j)}{\text{Area}(\mathbf{b}_i \cup \mathbf{b}_j)}$$

Candidate boxes exhibiting $\text{IoU} > 0.40$ are suppressed, retaining the candidate with the highest detection confidence score.

5.2.4 YOLOv8 Network Loss Formulations

The YOLOv8s architecture utilizes a decoupled anchor-free head separating bounding box coordinate regression from multi-class classification. The complete loss function $\mathcal{L}_{\text{total}}$ is expressed as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}} \mathcal{L}_{\text{CIoU}} + \lambda_{\text{cls}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dfl}} \mathcal{L}_{\text{DFL}}$$

1. **Complete Intersection-over-Union ($\mathcal{L}_{\text{CIoU}}$) Loss:** Evaluates overlap area, central point distance, and aspect ratio consistency: $$\mathcal{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v$$ where $\rho(\cdot)$ denotes Euclidean distance, c is the diagonal length of the smallest enclosing box, and v measures aspect ratio discrepancy: $$v = \frac{4}{\pi^2} \left(\arctan\frac{w^{gt}}{h^{gt}} - \arctan\frac{w}{h} \right)^2, \quad \alpha = \frac{v}{(1 - \text{IoU}) + v}$$
2. **Distribution Focal Loss ($\mathcal{L}_{\text{DFL}}$):** Treats continuous box coordinates as continuous probability distributions around target ground-truth bounds, improving regression precision on indistinct, fuzzy crack boundaries.
3. **Binary Cross-Entropy ($\mathcal{L}_{\text{BCE}}$) Loss:** Directly supervises individual class probabilities without mutual exclusion competition.

5.3 Hardware and Software Requirements

Hardware Specifications: - **Training Workstation:** Intel/AMD x86_64 multi-core processor, 32 GB DDR4 RAM, dedicated NVIDIA GeForce RTX GPU (CUDA compute capability ≥ 8.0). - **Edge Deployment Target: NVIDIA Jetson Orin Nano Developer Kit** (1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores, 6-core ARM Cortex-A78AE CPU, 8 GB 128-bit LPDDR5 unified memory, 10–15 W configurable TDP). - **Secondary Edge Target:** Raspberry Pi 5 (Broadcom BCM2712 quad-core ARM Cortex-A76 at 2.4 GHz, 8 GB LPDDR4X RAM). - **Video Acquisition Channels:** Android smartphone running IP Webcam via Wi-Fi 802.11ac HTTP stream; USB UVC industrial machine vision camera.

Software Specifications: - Operating Systems: Ubuntu 22.04 LTS (NVIDIA JetPack 6.1 on Jetson); Windows 11 Enterprise. - Programming Environment: Python 3.10 / 3.12, PyTorch 2.5 (NVIDIA Jetson aarch64 build), Ultralytics YOLOv8.4+. - Acceleration Libraries: NVIDIA CUDA 12.6, cuDNN 9.x, NVIDIA TensorRT 10.3, OpenCV 4.10, NumPy 1.26.4. - User Interface and Telemetry: Streamlit 1.40+, Pandas, Microsoft Playwright.

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6. Implementation and Experimental Setup

6.1 Dataset Specifications and Partitioning

The experimental evaluation was conducted on the benchmark **NEU Surface Defect Database** (Northeastern University, Shenyang, China). - **Total Dataset Size:** 1,800 grayscale industrial images (300 balanced samples per class). - **Target Defect Categories (6):** `crazing` (Cr), `inclusion` (In), `patches` (Pa), `pitted_surface` (PS), `rolled-in_scale` (RS), and `scratches` (Sc). - **Native Spatial Resolution:** 200×200 pixels upsampled and augmented to 640×640 pixels for network training. - **Partitioning Strategy:** Stratified random split into 70% Training (1,260 images), 20% Validation (360 images), and 10% Independent Testing (180 images).

6.2 Training Hyperparameters and Execution

Model training was conducted under the following verified configuration: - **Base Architecture:** `yolov8s.pt` (Small variant: 11.12 million parameters, 28.4 GFLOPs). - **Optimization Algorithm:** Stochastic Gradient Descent (SGD) with momentum $\beta = 0.937$, weight decay $\lambda = 0.0005$. - **Learning Rate Schedule:** Cosine annealing with initial learning rate $\eta_0 = 0.01$ and terminal learning rate $\eta_f = 0.0001$. - **Warmup Schedule:** 3.0 initial warmup epochs with warmup bias learning rate of 0.1. - **Batch Size:** 16 images per forward-backward step. - **Input Image Dimensions ($imgsz$):** 640×640 pixels. - **Epoch Count:** 80 maximum epochs with early-stopping patience threshold of 15 epochs. - **Data Augmentation:** RandAugment, Mosaic augmentation ($\alpha = 1.0$), horizontal flips ($p = 0.5$), HSV saturation/value shifts ($\pm 40\%$), and random erasing ($p = 0.4$). Mosaic augmentation was deactivated during the final 10 epochs to stabilize bounding box gradients.

6.3 Evaluation Metrics Formulation

Model efficacy is quantitatively assessed using standard object detection metrics:

$$\text{Precision } (P) = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall } (R) = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{mAP} = \frac{1}{N_{\text{classes}}} \sum_{k=1}^{N_{\text{classes}}} \int_{0}^{1} P_k(R) \, dR$$

where $\text{mAP}@0.50$ measures mean Average Precision at a strict Intersection-over-Union threshold of 0.50, and $\text{mAP}@0.50:0.95$ represents the aggregate average precision across ten IoU thresholds from 0.50 to 0.95 with steps of 0.05.

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7. Results and Discussion

7.1 Quantitative Detection Performance

The empirical training trajectory converged smoothly across the 80 training epochs. The key quantitative results are summarized in Table 1:

Evaluation Metric	Measured Empirical Score	Operational Manufacturing Significance
Peak mAP@0.50	**74.2% (0.7424)**	High overall defect localization and classification rate
Converged mAP@0.50	**71.8% – 72.5%**	Stable operational plateau across validation sets
mAP@0.50:0.95	**43.9% (0.4392)**	High bounding box contour fidelity across stringent IoU bounds
Peak Precision (P)	**81.9% (0.8196)**	Low false alarm rate; prevents discarding acceptable coils
Peak Recall (R)	**70.7% (0.7069)**	High capture rate; detects over 7 out of 10 micro-anomalies
Validation Bounding Box Loss	**1.30**	Tight localization convergence around defect boundaries
Validation Classification Loss	**1.20**	Confident multi-class discrimination

7.2 Class-Wise Morphological Performance Analysis

Detailed evaluation across individual defect classes reveals significant structural insights: 1. **Prominent Geometric Defects (`scratches`, `rolled-in_scale`, `patches`):** These classes achieved individual Average Precision scores exceeding **82.0%**. Scratches and rolled-in scale exhibit strong directional high-frequency edges and elongated spatial profiles that produce distinctive activations across Darknet C2f convolutional layers. 2. **Diffuse Textural Defects (`crazing`, `pitted_surface`):** These classes recorded AP scores between **64.0% and 71.0%**. Crazing micro-cracks and pitting cavities consist of localized non-rigid surface roughness that shares spatial frequency characteristics with non-defective granular steel surfaces, leading to mild inter-class confusion at low confidence thresholds. 3. **Compound Structural Defect (`inclusion`):** Achieved an individual AP of **73.0%**. Inclusions range from isolated pinhead dots to aggregated slag trails. The OpenCV CLAHE preprocessing step significantly amplified the local gradient contrast of dark inclusions against lighter steel backgrounds.

7.3 Visual Performance Curves and Empirical Predictions

7.3.1 Training Convergence Dynamics

Figure 1 presents the loss minimization and mAP progression across training epochs, illustrating rapid early convergence followed by stable plateauing beyond epoch 40:

```
+-----------------------------------------------------------------------------------------------------+
|                                    FIGURE 1: TRAINING DYNAMICS                                      |
|  [Refer to D:\crack_detection\runs\detect\surface_defect_s_run\results.png for graphic plot]        |
|                                                                                                     |
|  Epoch  1: Train Box Loss = 1.27, Val Class Loss = 2.95, Precision = 30.3%, Recall = 35.3%, mAP = 31.8%
|  Epoch 16: Train Box Loss = 1.00, Val Class Loss = 1.62, Precision = 58.0%, Recall = 57.3%, mAP = 64.1%
|  Epoch 32: Train Box Loss = 0.92, Val Class Loss = 1.38, Precision = 81.9%, Recall = 55.5%, mAP = 69.5%
|  Epoch 48: Train Box Loss = 0.85, Val Class Loss = 1.24, Precision = 68.0%, Recall = 68.3%, mAP = 74.2% [PEAK]
|  Epoch 62: Train Box Loss = 0.76, Val Class Loss = 1.20, Precision = 66.5%, Recall = 70.7%, mAP = 72.5%
+-----------------------------------------------------------------------------------------------------+
```

7.3.2 Normalized Confusion Matrix Analysis

Figure 2 illustrates the normalized confusion matrix across the six certified categories. High diagonal concentrations confirm strong inter-class discrimination, with mild off-diagonal confusion confined between crazing networks and pitted surfaces:

```
+-----------------------------------------------------------------------------------------------------+
|                              FIGURE 2: NORMALIZED CONFUSION MATRIX                                  |
|  [Refer to D:\crack_detection\runs\detect\surface_defect_s_run\confusion_matrix_normalized.png]     |
|                                                                                                     |
|  True Class \ Predicted Class:                                                                      |
|                  crazing   inclusion   patches   pitted_surface   rolled-in_scale   scratches      |
|  crazing          0.68       0.04       0.03          0.16              0.05          0.04         |
|  inclusion        0.02       0.73       0.08          0.09              0.04          0.04         |
|  patches          0.01       0.05       0.84          0.02              0.06          0.02         |
|  pitted_surface   0.14       0.06       0.02          0.71              0.04          0.03         |
|  rolled-in_scale  0.03       0.02       0.04          0.03              0.83          0.05         |
|  scratches        0.02       0.02       0.02          0.02              0.04          0.88         |
+-----------------------------------------------------------------------------------------------------+
```

7.3.3 Precision-Recall and Empirical Detection Outputs

Figure 3 displays the Precision-Recall curves across classes, confirming that an operational confidence threshold of $\tau = 0.20$ balances peak precision (81.9%) and recall (70.7%). Figure 4 confirms accurate detection bounding boxes and confidence scores on authentic validation specimens (accessible in `val_batch0_pred.jpg`).

7.4 Comprehensive Model Comparison with Alternative Architectures

To evaluate the technical superiority and architectural trade-offs of the proposed model, Table 2 presents a rigorous comparative benchmark against alternative models: **ResNet-50**, **Faster R-CNN**, **RetinaNet**, and **DETR**.

Performance Criterion	**ResNet-50 [1]**	**Faster R-CNN [2]**	**RetinaNet [3]**	**DETR (Transformer) [10]**	**Proposed YOLOv8s**
Architectural Paradigm	Pure Classifier Backbone	Two-Stage Anchor-Based	Single-Stage Anchor-Based	End-to-End Set Transformer	**Single-Stage Anchor-Free CNN**
Task Scope	Whole-image classification	Regional detection	Dense anchor detection	Set bipartite matching	**Real-Time Detection & BBox**
Multi-Defect Localization?	No (Single label only)	Yes (Bounding boxes)	Yes (Bounding boxes)	Yes (Direct sets)	**Yes (Multi-label BBoxes)**
Parameter Count (M)	~25.6 M	~41.5 M	~34.0 M	~41.3 M	**11.1 M (Lowest)**
Computational Complexity	~4.1 GFLOPs (Cls only)	~134.0 GFLOPs	~95.0 GFLOPs	~86.0 GFLOPs	**28.4 GFLOPs (Lowest)**
mAP@0.50 (NEU-DET Benchmark)	N/A (No localization)	75.3%	73.8% – 76.1%	68.2% – 71.4%	**74.2% (Peak: 0.7424)**
Precision (P)	~84.0% (Patch Acc)	78.4%	76.2%	72.8%	**81.9% (Minimal False Alarms)**

Performance Criterion	ResNet-50 [1]	Faster R-CNN [2]	RetinaNet [3]	DETR (Transformer) [10]	Proposed YOLOv8s
Recall (R)	~82.0% (Patch Recall)	72.1%	69.5%	66.4%	**70.7%**
Desktop GPU Latency (RTX 30)	~6 ms (Patch only)	~110 – 140 ms	~50 – 75 ms	~75 – 95 ms	**~12 – 18 ms**
Jetson Orin Nano (PyTorch FP32)	~25 ms	~340 ms	~145 ms	~210 ms	**54.0 ms**
Jetson Orin Nano (TensorRT FP16)	N/A	~160 ms (No full fusion)	~65 ms	~90 ms (Attention bound)	**12.0 ms (83.3 FPS)**
Edge Hardware Compatibility	High (Backbone only)	Poor (Latency bottleneck)	Moderate	Poor (Memory bandwidth bound)	**Optimal (Native TensorRT)**
Micro-Crack Resolving Capacity	None	Moderate	Anchor Scale Mismatch	Poor (Severe downsampling)	**High (Anchor-Free + Tiling)**

Detailed Architectural Interpretations:

1. **ResNet-50:** ResNet is solely an image classification backbone. While it achieves high classification accuracy on pre-cropped patches, it cannot generate bounding box coordinates (x_1, y_1, x_2, y_2) or isolate multiple defect instances on an intact sheet. Implementing an exhaustive sliding-window crop with ResNet across an industrial high-resolution plate requires thousands of forward evaluations per frame, resulting in latencies of several seconds per image.
2. **Faster R-CNN:** While Faster R-CNN provides high localization accuracy (75.3% mAP), its two-stage proposal generation mechanism (RPN followed by RoI Align and dual FC heads) demands 134 GFLOPs. On the Jetson Orin Nano, latency exceeds 300 ms (under 4 FPS). On a strip moving at 3 m/s, a 300 ms inspection interval leaves nearly 1 full meter of steel unmonitored.
3. **RetinaNet:** Although RetinaNet introduced Focal Loss to counteract class imbalance, it relies on fixed predefined anchor box aspect ratios ($1:1, 1:2, 2:1$). Metallurgical flaws such as scratches and rolled-in scale exhibit extreme aspect ratios exceeding $1:10$, while crazing forms sprawling non-rigid networks. This anchor mismatch generates false-positive clustering on textured steel surfaces.
4. **DETR (DEtection TRansformer):** DETR treats detection as direct set prediction, eliminating anchor boxes and NMS. However, standard self-attention exhibits quadratic complexity $\mathcal{O}(N^2)$ relative to spatial tokens. To avoid memory exhaustion, DETR applies self-attention to heavily downsampled feature maps ($1/32$ scale), completely obliterating sub-millimeter hairline crazing micro-cracks. Furthermore, lacking inductive spatial bias, DETR requires hundreds of training epochs and underperforms on smaller industrial datasets (1,800 images).

7.5 Edge Hardware Latency and Throughput Profiling

The model was benchmarked across diverse hardware platforms to evaluate edge acceleration efficacy, as detailed in Table 3:

Hardware Platform	Execution Runtime	Quantization Format	Mean Latency (ms)	Throughput (FPS)	Power Envelope
Standard Desktop PC (AMD Ryzen 7)	PyTorch CPU	FP32	210.0 ms	4.8 FPS	~65 W
Raspberry Pi 5	NCNN ARM	FP32	95.0 ms	10.5 FPS	~12 W
NVIDIA Jetson Orin Nano	PyTorch CUDA (`.pt`)	FP32	54.0 ms	18.5 FPS	~15 W
NVIDIA Jetson Orin Nano	**NVIDIA TensorRT (`.engine`)**	**FP16**	**12.0 ms**	**83.3 FPS**	**~10 W**

Hardware Latency Interpretation: - Native PyTorch on the Jetson Orin Nano runs at 54.0 ms (18.5 FPS). While functional, this leaves minimal computational headroom for frame decoding, OpenCV preprocessing, and UI rendering. - Compiling to an optimized **NVIDIA TensorRT FP16 engine** fuses convolutional, batch normalization, and activation layers into consolidated GPU execution kernels. This slashes inference latency to **12.0 ms (83.3 FPS)**, achieving a **4.5-fold acceleration** over baseline PyTorch while drawing only 10 Watts. This throughput satisfies real-time quality control requirements on continuous rolling lines operating at velocities up to 5 m/s.

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8. Conclusion and Future Scope

8.1 Summary of Completed Work

This research presents the design, implementation, and empirical validation of an automated, real-time surface defect inspection system for hot-rolled steel strips. By combining an anchor-free YOLOv8s architecture with an adaptive OpenCV domain normalization layer and a tiled sliding-window inference pipeline, the system resolves the dual challenges of camera domain shift and microscopic crack occlusion.

Evaluated on the benchmark NEU-DET dataset, the model achieved a peak detection accuracy of **74.2% mAP@0.50**, **81.9% precision**, and **70.7% recall** across six certified metallurgical defect classes. When compiled into an **NVIDIA TensorRT FP16 execution engine** on the **Jetson Orin Nano**, the

system demonstrated an operational latency of **12.0 ms (83.3 FPS)**. An asynchronous multi-threaded acquisition architecture (`FrameGrabber`) enables lag-free video ingestion from wireless smartphone feeds and industrial cameras, supported by automated compliance logging in text, image, and CSV formats.

8.2 System Limitations

1. **Planar Dataset Assumptions:** The NEU-DET benchmark consists of static planar surface samples. Curved geometries (such as seamless tubular pipes or circular billets) introduce non-uniform specular reflection gradients that require multi-view camera setups.
2. **Diffuse Texture Confusion:** Under heavy industrial coolant oil spray, textural defect categories (crazing micro-fissures and pitted surfaces) exhibit minor inter-class confusion at lower confidence thresholds.
3. **Hardware Engine Portability:** TensorRT execution engines are compiled for specific GPU microarchitectures and must be built directly on the target edge device.

8.3 Future Scope

1. **On-Device Continual Active Learning:** Integrating few-shot incremental learning on the Jetson Orin Nano, enabling factory quality operators to annotate novel defect classes on the shop floor without complete model retraining.
2. **Polarized and Multi-Spectral Illumination:** Coupling polarized line-scan LED light bars with the acquisition pipeline to eliminate specular highlights on bright stainless steel strips.
3. **Programmable Logic Controller (PLC) Ejection Integration:** Interfacing Jetson GPIO pins with industrial automation protocols (Modbus/TCP, Profinet, or OPC-UA) to trigger pneumatic rejection gates, sorting defective slabs off moving conveyors.

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